Physiological Changes in Pregnancy

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Dr. Ojike – Obstetrics & Gynaecology, topic 6, Wed 27/05/2026

Introduction

Pregnancy represents a **unique physiological state**, characterised by **profound, reversible adaptations across virtually every organ system** in the maternal body.

These adaptations are **not pathological**. Rather, they are **highly coordinated responses** designed to:

1. **Support the metabolic demands** of the growing fetus and placenta
2. **Protect the mother** from the physiological stressors of pregnancy and parturition (especially **haemorrhage during delivery**)
3. **Ensure adequate utero-placental perfusion** for nutrient and gas exchange

These changes **start after conception** and are driven primarily by **endocrine signals from the corpus luteum and the developing placenta** – notably **oestrogen, progesterone and human chorionic gonadotropin (hCG)**.

Cardiovascular system

The cardiovascular system adaptation ensures that **oxygen and nutrients are delivered to the uterus and placenta**.

The elevated oestrogen and progesterone cause:

- **Systemic vascular resistance to be lowered**
- **Cardiac output to increase**

Blood volume changes

- **Plasma volume increases by 40-50%** above the non-pregnant level
- **Red blood cell (RBC) mass increases by only 15-20%**, stimulated by increased **erythropoietin** production

Physiological anaemia of pregnancy

Plasma volume expansion far outpaces the rise in RBC mass

decreased

Haemodilution – a fall in haemoglobin concentration and haematocrit

> **Clinical importance:** in pregnancy, haemoglobin levels of **10.5 g/dL - 11 g/dL** are considered **physiologically normal**.

Cardiac output

Cardiac output increases by 30-50% in pregnancy.

Cardiac output (CO) = Heart Rate (HR) x Stroke Volume (SV)

- **Stroke volume increases early in pregnancy (20-30%)**, due to **increased end-diastolic volume and myocardial hypertrophy**
- **Heart rate increases progressively throughout pregnancy (15-20 beats per minute above baseline)**

Systemic vascular resistance and blood pressure

SVR decreases (25-30%), driven by:

1. The **smooth muscle relaxing effect of progesterone**
2. **Increased local production of vasodilators** – **nitric oxide (NO)** and **prostacyclin (PGI2)**
3. The **creation of a low-resistance, high-flow utero-placental circuit**

Blood pressure (BP): despite the increase in cardiac output, BP **decreases** due to the massive drop in SVR.

- The **diastolic drops more than the systolic**
- The **lowest BP is reached around 20-24 weeks**, before gradually returning to the pre-pregnancy baseline at term

> **N.B.** This is the reason for using **after 20 weeks gestation as the cut-off** for gestational hypertension (PIH).

Aortocaval compression

N.B. When a pregnant woman lies **supine**, the gravid uterus compresses the **inferior vena cava (IVC)** and the **lower aorta**, leading to **reduced venous return to the heart**

=> **drop in stroke volume and cardiac output**

Supine hypotensive syndrome – dizziness, pallor and placental hypoperfusion

Hence pregnant women in late pregnancy should **always lie in left lateral tilt**, to displace the uterus off the great vessels.

Respiratory system

The respiratory system adapts to ensure that the **increased metabolic demand for oxygen – elevated by roughly 20%** – is met, and that **fetal waste carbon dioxide is effectively cleared**.

- **Tidal volume increases (30-40%)**
- **Functional residual capacity is reduced (10-20%)**
- **Minute ventilation** (tidal volume x respiratory rate) **increases by 30-40%**
- **Residual volume decreases by 15-20%**
- **Vital capacity – unchanged**

Renal system

The maternal kidneys must process **both maternal waste and fetal metabolic byproducts**.

Anatomical changes

Hydronephrosis and hydroureters, due to:

1. **Progesterone-induced smooth muscle relaxation**
2. **Compression of the ureters at the pelvic brim by the expanding uterus** – more on the **right** than on the **left**, due to:
   - The **dextrorotation of the uterus**
   - The **cushioning effect of the left ureter by the sigmoid colon**

N.B. The **stasis of urine** increases the risk of progression from **asymptomatic bacteriuria** to **acute pyelonephritis**.

Renal haemodynamics

- **Renal plasma flow increases by up to 50-80%**
- **Glomerular filtration rate (GFR) increases by up to 50%**

N.B. Hyperfiltration => clearance of **creatinine, urea and uric acid** is highly elevated

=> **decreased serum creatinine (0.4-0.6 mg/dL)** against a non-pregnant level of **0.8 mg/dL**

Hence a **creatinine level of 0.8 mg/dL or higher in pregnancy warrants evaluation** for underlying **renal impairment or pre-eclampsia**.

- **High GFR leads to exceeding the renal threshold for glucose complete reabsorption | Glycosuria**
- There is also a **mild increase in urinary protein excretion**

Gastrointestinal system

The GI changes of pregnancy are **primarily functional**, driven by the **smooth muscle relaxant properties of progesterone**.

High levels of progesterone lead to:

- **decreased lower oesophageal sphincter (LES) tone => GERD (gastro-oesophageal reflux disease)**, caused by decreased LES tone **combined with increased intra-abdominal pressure from the expanding uterus**
- **Decreased intestinal motility**
- **Decreased gall bladder motility => cholelithiasis**
- **Constipation**

Oestrogen causes **gingival hyperplasia and bleeding gums (epulis gravidarum)**.

Haemorrhoids – caused by **constipation** and **elevated venous pressure in the pelvic veins** due to uterine compression of the vena cava.

Alkaline phosphatase (ALP) doubles or triples, due to **placental production**.

Haematological system

Pregnancy is characterised by a **hypercoagulable state** – an **evolutionary adaptation to prevent fatal postpartum haemorrhage**, but one that carries **significant risk of venous thromboembolism (VTE)**.

- **Increases in procoagulant factors – fibrinogen, VII, VIII, IX and XII** – as well as **von Willebrand factor**
- **Decrease in the anticoagulant protein S**
- **Fibrinolysis decreased**

  There is an increased risk of **deep vein thrombosis (DVT)** and **pulmonary embolism (PE) – up to 4-5 fold**.

- **Leucocytosis**

Endocrine system

The endocrine milieu undergoes a **complete revolution**, transforming maternal metabolism to **prioritise the continuous supply of glucose and amino acids to the fetus**.

increased Oestrogen, human placental lactogen, progesterone, cortisol, placental growth hormone

Hormone-induced insulin resistance => the maternal response must **increase insulin secretion to maintain euglycaemia**

> **N.B. Gestational diabetes mellitus** occurs when the **maternal pancreatic response is insufficient to overcome this hormone-induced insulin resistance**.

Pituitary gland

- The **pituitary gland enlarges up to 135%**, due to **primary hypertrophy and hyperplasia of lactotrophs (prolactin-producing cells)**, stimulated by **oestrogen** |

- **Prone to Sheehan syndrome** if severe **post-partum haemorrhage and hypovolaemic shock** occur

Integumentary / musculoskeletal system

- **Hyperpigmentation**, due to elevated **MSH (melanocyte stimulating hormone)** and **oestrogen**

  This leads to **darkening of the areolae, the perineum, axillae, and the midline of the abdomen (linea nigra)**

- **Melasma (chloasma)**
- **Striae gravidarum**
- **Vascular changes – palmar erythema, spider angiomas**
- **Progressive lumbar lordosis**
- **Joint laxity**